

Bank Customer Analytics & AI Churn Prediction Dashboard

Beginner-Friendly Python + AI Project

1. Introduction

This project demonstrates how Python and basic Artificial Intelligence can be used to analyze banking customer data and predict whether a customer may be at risk of churn. The project is designed for beginners and can be presented as an academic or portfolio project.

2. Objectives

- Analyze customer age, income, account balance, transactions and credit score.
- Calculate simple banking KPIs.
- Build an interactive dashboard using Streamlit.
- Train a simple Decision Tree machine-learning model.
- Predict low or high churn risk for a new customer.

3. Tools & Technologies

Python, Pandas, NumPy, Scikit-learn and Streamlit.

4. Dataset

The project uses a synthetic dataset of 500 customers. It contains Customer ID, Age, Monthly Income, Account Balance, Monthly Transactions, Credit Score, Loan status and Churn. The data is synthetic and is not real customer or bank data.

5. Methodology

Step 1: Load the CSV dataset. Step 2: Select useful features. Step 3: Split the data into training and testing sets. Step 4: Train a Decision Tree Classifier. Step 5: Enter a new customer's information in the dashboard. Step 6: Display the predicted churn risk and a simple business explanation.

6. Dashboard

The dashboard displays total customers, average income, average account balance, churn rate, customer analytics charts, sample customer records, and an AI prediction panel.

7. AI Model

A Decision Tree Classifier is used because it is beginner-friendly and easy to explain. The model learns patterns from the training data and predicts whether a customer belongs to the churn or non-churn class.

8. Business Recommendation

A bank could use a similar approach to identify customers who may require additional attention. In a real organization, predictions should be validated carefully and should not be used as the sole basis for customer decisions.

9. Conclusion

The project combines data analytics, machine learning and an interactive dashboard in a simple banking use case. It demonstrates practical Python skills and basic AI understanding.

10. Viva Explanation

“I created a banking customer analytics project using Python. I used Pandas for data handling, Streamlit for the interactive dashboard and a Decision Tree model from Scikit-learn for churn prediction. The dashboard allows a user to enter customer information and receive a simple churn-risk prediction.”